Economic History II

Media, propaganda and opinion

FGV EPGE

Outline of the class

- **DellaVigna, Enikolopov, Mironova, Petrova and Zhuravskaya (2014)**, the paper you read: Serbian radio spilling across the border into Croatia.
- Adena, Enikolopov, Petrova, Santarosa and Zhuravskaya (2015): radio in Weimar and Nazi Germany.
- Same tool in both: signal propagation and terrain decide who hears what.
- Different regimes, different content, different listeners.

Serbian radio in Croatia

DellaVigna, Enikolopov, Mironova, Petrova and Zhuravskaya (2014)

Cross-border media and nationalism:

Evidence from Serbian radio in Croatia [⊥]

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August 2013

Abstract

How do nationalistic media affect animosity between ethnic groups? We consider one of Europe's deadliest conflicts since WWII: the Serbo-Croatian conflict. We show that, after a decade of peace, cross-border nationalistic Serbian radio triggers ethnic hatred towards Serbs in Croatia. Mostly attracted by non-political content, many Croats listen to Serbian public radio (intended for Serbs in Serbia) whenever signal is available. As a result, the vote for extreme nationalist parties is higher, and ethnically offensive graffiti are more common, in Croatian villages with Serbian radio reception. A laboratory experiment confirms that Serbian radio exposure causes anti-Serbian sentiment among Croats.

Recall: state and nation

- Gellner: nationalism holds that "the political and the national unit should be congruent".
- Almost nowhere true.
- Two routes: move the borders to fit the people, or remake the people to fit the borders.



The ethnic groups of Austria-Hungary in 1910 according to *Distribution of Races in Austria-Hungary* by William R. Shepherd, 1911.

Serbo-Croatian

112 languages

Article Talk Read View source View history

From Wikipedia, the free encyclopedia

(Redirected from Serbo-Croatian language)

This article is about the language. For other uses, see Serbo-Croatian (disambiguation).

Serbo-Croatian,[b] also known as **Bosnian-Croatian-Montenegrin-Serbian** (**BCMS**),[11][c] is a Western South Slavic language and the primary language of Serbia, Croatia, Bosnia and Herzegovina, and Montenegro.[14] It is a pluricentric language with four[15] mutually intelligible standard varieties, namely Serbian, Croatian, Bosnian, and Montenegrin.[16][14]

The western Balkans' turbulent history, particularly due to the expansion of the Ottoman Empire, led to a complex dialectal and religious mosaic. Due to population migrations, Shtokavian became the most widespread supradialect in the region, encroaching westward into the area previously dominated by Chakavian and Kajkavian.[17] Bosniaks, Croats, and Serbs differ in religion and were historically often part of different cultural spheres, although large portions of these populations lived side by side under foreign rule. During that period, the language was referred to by various names, such as "Slavic" in general, or "Serbian", "Croatian" or "Bosnian" in particular. In a classicizing manner, it was also referred to as "Illyrian".

Serbo-Croatian	
Bosnian-Croatian-Montenegrin-Serbian	
srpskohrvatski / српскохрватски	
Pronunciation	[sr̩pskoxr̝ʊaːtskiː]
Native to	Serbia · Croatia · Bosnia and Herzegovina · Montenegro · Kosovo
Ethnicity	Serbs · Croats · Bosniaks · Montenegrins
Native speakers	17 million (2020–2023) [1]
Language family	Indo-European • Balto-Slavic • Slavic • South Slavic • Western South Slavic • Serbo-Croatian

Sociolinguistic debate

The nature and classification of Serbo-Croatian has been the subject of long-standing sociolinguistic debate.[103] The question is whether Serbo-Croatian should be called a single language or a cluster of closely related languages.[104][12][105][106]

Views of linguists in the former Yugoslavia

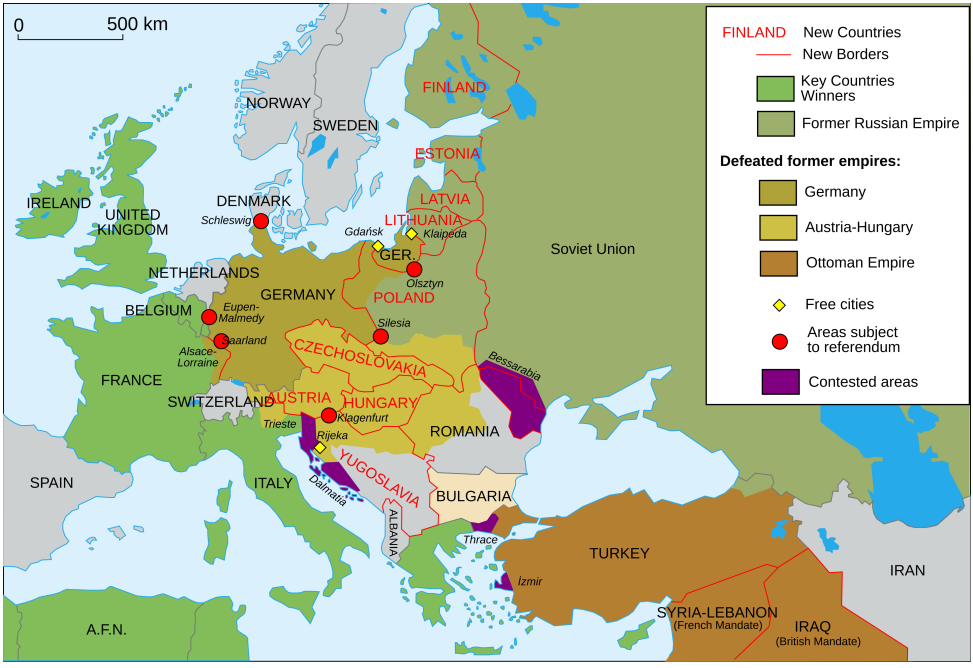
Views of Croatian linguists

A prevailing view among Croatian linguists is that there has never been a Serbo-Croatian language, but rather two different standard languages that overlapped sometime in the course of history. However, Croatian linguist Snježana Kordić led an academic discussion[107] on this issue in the Croatian journal *Književna republika*[108] from 2001 to 2010.[109][110] The editor of that magazine says: "I encouraged prominent Croatian linguists to respond to her. However, after a while it became

Ethnicity and nation

- Max Weber: an ethnic group holds "a subjective belief in their common descent", resting on customs, physical type or memories of migration. Whether the blood tie is real does not matter.
- Political communities can themselves inspire this belief.
- A nation draws on the same belief and adds a political aim: an autonomous polity, a state of its own.
- A common language is not enough. Weber's own example: Serbs and Croats, closely related by descent, are different nationalities because of their religion.

Weber, *Economy and Society*, pp. 389, 395. Screenshot: Wikipedia, "Serbo-Croatian".

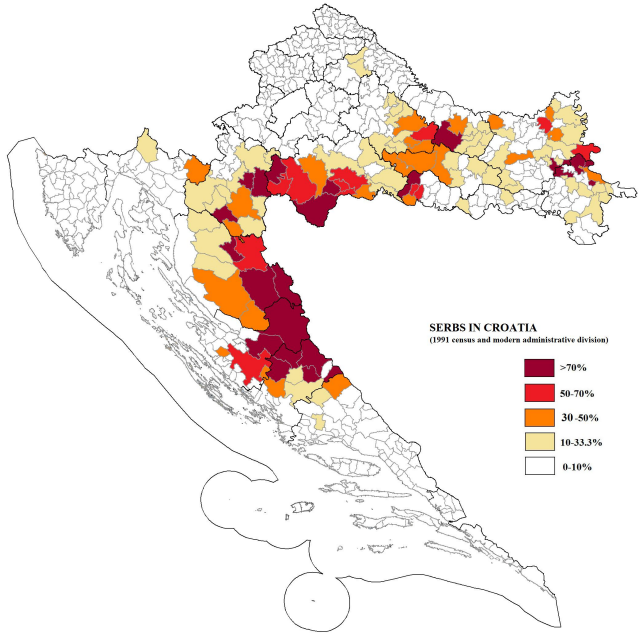

0
500 km
FINLAND New Countries
New Borders
Key Countries Winners
Former Russian Empire
Defeated former empires:
Germany
Austria-Hungary
Ottoman Empire
Free cities
Areas subject to referendum
Contested areas
NORWAY
SWEDEN
FINLAND
ESTONIA
LATVIA
LITHUANIA
Klaipeda
Gdańsk
GER
Olsztyn
POLAND
Silesia
Soviet Union
IRELAND
UNITED KINGDOM
DENMARK
Schleswig
NETHERLANDS
GERMANY
BELGIUM
Eupen-Malmedy
Saarland
Alsace-Lorraine
FRANCE
SWITZERLAND
CZECHOSLOVAKIA
Bessarabia
AUSTRIA
HUNGARY
Klagenfurt
Trieste
Rijeka
ROMANIA
SPAIN
ITALY
YUGOSLAVIA
Dalmatia
BULGARIA
ALBANIA
Thrace
İzmir
TURKEY
IRAN
SYRIA-LEBANON
(French Mandate)
IRAQ
(British Mandate)
A.F.N.

Yugoslavia: one route, then the other

- 1918: Kingdom of Serbs, Croats and Slovenes, Yugoslavia from 1929. Borders moved around most South Slavs.
- Then the second route: one state trying to hold several nations together.
- It broke apart in the 1990s.

Map: Fluteflute and Historicair, Wikimedia Commons, CC BY-SA 2.5.


Former Yugoslavia
Austria
Hungary
Slovenia
LJUBLJANA
ZAGREB
Croatia
Vojvodina (autonomous province)
Romania
Belgrade
Bosnia and Hercegovina
SARAJEVO
Serbia
Adriatic Sea
Italy
Montenegro
Kosovo (autonomous province)
Bulgaria
Macedonia
Albania
Greece
Hungarian, 1.9
Other, 3.9
Montenegrin, 2.5
Serb, 36.3
Yugoslav, 5.4*
Macedonian, 5.9
Albanian, 7.7
Slovene, 7.8
Muslim, 8.9
Croat, 19.7
*Yugoslavs are those persons who listed themselves as such in the 1981 census. They are dispersed across the country.
Albanian
Bulgarian
Croat
Hungarian
Macedonian
Montenegrin
Muslim
Serb
Slovak
Slovene
No majority present
Based on opstina data from 1991 census.
0
100 Kilometers
0
100 Miles
725588 (R00472) 6-92


SERBS IN CROATIA
(1991 census and modern administrative division)
>70%
50-70%
30-50%
10-33.3%
0-10%

1991: the mismatch again

- Croats want their own state.
- The Serbian minority inside it, backed by Serbia, wants to stay in Yugoslavia.
- Each side's nation-state leaves someone on the wrong side of the border.
- The third route: **move the people**. Ethnic cleansing on both sides, roughly 250,000 displaced.

Maps: CIA, *The Former Yugoslavia: A Map Folio* (1992), public domain; MirkoS18, Wikimedia Commons, CC BY-SA 3.0.

After 1995: one language, two states

- Croatia independent, within the borders it claimed in 1991.
- Two states now building two nations next door to each other.
- Gellner: the state's monopoly of legitimate education. Schools stop at the border.
- **Radio does not.** Serbian public radio's mission is "to strengthen Serbian national identity", and Croats can fully understand it.

The question

- Which factors fuel animosity between ethnic groups?
- Here: media meant to strengthen the identity of one group, heard by a rival group.
- Radio and television waves travel across political and ethnic boundaries.
- Common where neighbors share a language: the former Yugoslavia, the former USSR, Sudan.
- Also within countries: Muslims and Hindus in India, Basques and Spanish in Spain.

Three answers are possible a priori

- **Animosity rises**: views polarize, future conflict becomes more likely.
- Or exposure reduces informational asymmetries between groups and eases tension.
- Or the cross-group effect is negligible: people mostly consume media that support their own views.
- The literature had studied propaganda on its *intended* audience. Not on the audience across the border.

The conflict

- Croatian-Serbian conflict, 1991 to 1995, part of the Yugoslavian wars.
- Croats predominantly Catholic, Serbs predominantly Orthodox. Animosity escalating since the 1980s.
- Massacres of civilians and ethnic cleansing on both sides.
- Roughly 250,000 displaced civilians and 20,000 casualties, the vast majority civilians.
- Milosevic overthrown in 2000 and handed to the Hague International Criminal Tribunal.



- Eastern Slavonia, Baranja and Western Syrmia: Serbian Autonomous Oblast (1991), part of the Republic of Serbian Krajina (1991-1995), self-governing entity (1995-1996), UN-administered territory (1996-1998).

Note: borders presented in this map are from 1994.

The region

- Eastern Slavonia, Baranja and Western Srijem, on the border with Serbia.
- Site of full-scale armed conflict in 1991, under Serbian occupation until 1995.
- Two counties. Vukovar-Sirmium suffered most; all Croats there were displaced or killed during the Krajina period.
- Baranja has a large Hungarian minority and much weaker Croatian nationalism.
- The comparison is always **within county**.

Map: PANONIAN, Wikimedia Commons, public domain.

Why the setting works

- "The Serb-Croat case is a nearly ideal setting to study cross-group effects of media."
- Serbs and Croats speak the same language, with different alphabets.
- The gap is about the size of the gap between British and American English.
- Serbian public radio is made for listeners inside Serbia, and its signal reaches some Croatian villages and not others.

Why radio, and not television or newspapers

- Rural villages: low penetration of television sets and newspapers.
- Radio is cheap and easily accessible, the main political information source since Socialist Yugoslavia.
- A typical Croatian listens to radio about 250 minutes a day.
- Most people leave the radio on all day, and many listen on the street.
- Local stations dominate: the three national public channels hold a joint share of just 10% of radio audiences.

Serbian public radio

- Four RTS public stations: news, art, intellectual programs, and Radio Belgrade 202 (popular music).
- They “operate with the official mission to strengthen Serbian national identity”.
- 9% of coverage goes to neighboring countries, focused on war and war crimes there.
- **Turbo-folk**, invented under Milosevic, is considered offensive by many Croats and Bosnians: it glorifies Serbs during the war.
- One private exception: Radio B92, founded in 1989 by the Soros foundation and USAID.

Croats are an off-target audience

- The vast majority of listeners to these stations are Serbs in Serbia.
- If the target were Croats, the transmitters would sit closer to the border to push the signal in.
- So the paper identifies “what is likely to be an unintended media effect on an off-target audience as a catalyst of ethnic hatred”.

Two measures of reception

- **Hand-collected**: the authors drove through the region in June 2009 and June 2010 with a portable receiver.
- In the center of each village, near the church or the war monument, they tested whether RTS stations came in.
- Available if two people judged the quality good enough to listen to. 139 villages, 16 with reception.
- **Predicted signal strength**: RTS transmitter locations and power plus topography, through the Irregular Terrain Model.
- 417 villages within 75 kilometers of the border.

Figure 1a. Map of the area with the baseline and extended samples of villages

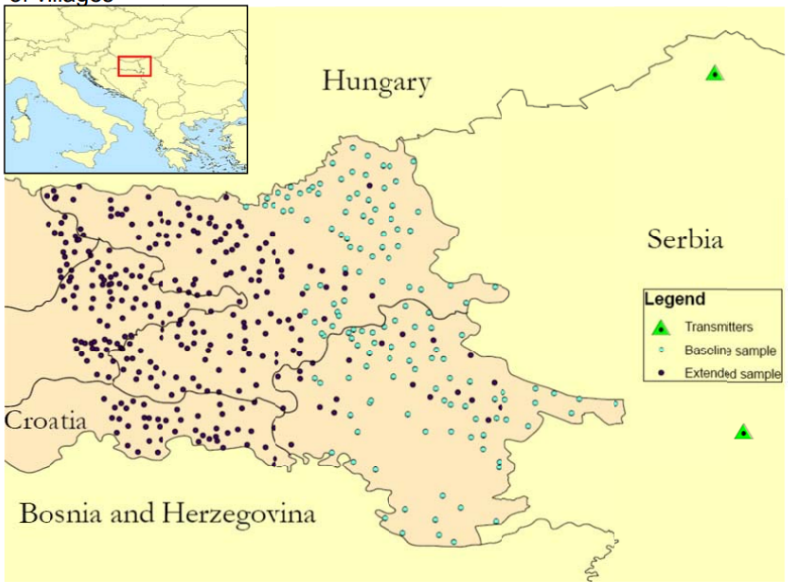
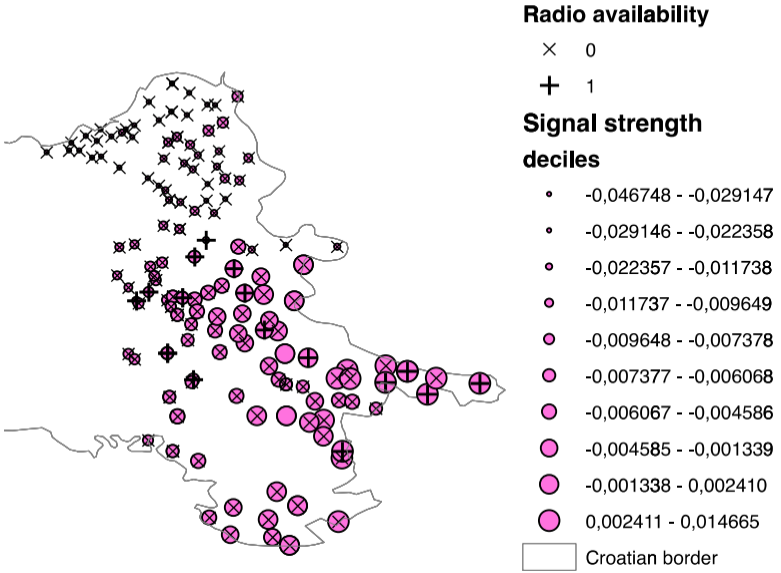


Figure 1b. Map of the area with the baseline sample of villages showing both the measured and the predicted reception of Serbian radio.



Measured reception against predicted signal

- Crosses and plus signs: RTS unavailable and available on the visit.
- Circle size: predicted signal strength, in deciles.
- Availability declines with distance from the Serbian border, but in a jagged way.
- **The jaggedness is the variation the paper uses.**

The parties

- HDZ took 36.6% of the national vote in 2007 and 66 seats; SDP 31.2% and 56 seats.
- HSP is the main extreme nationalist party, with the ideology of “the Great Croatia”.
- In the border region it is not marginal: 8% of the vote in the north, 5.2% in the south.
- In the 1990s HSP ran its own paramilitary unit, with symbols resembling those of the wartime fascist state.

First question

Do Croats actually listen?

The street survey

- December 2010 and January 2011: 70 people interviewed in 9 villages.
- 4 villages coded as having no Serbian radio, 5 with at least one station.
- Interviewers stopped people on the central street. Response rate about 50%.
- Small and not representative, so the paper treats it as suggestive.

Figure 2a. Reported reception of Serbian radio (survey response), as function of availability of Serbian radio in village.

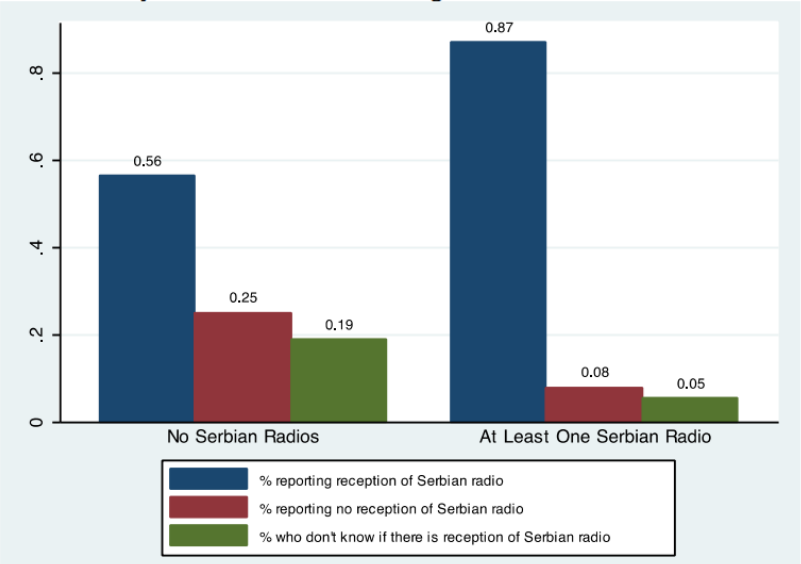
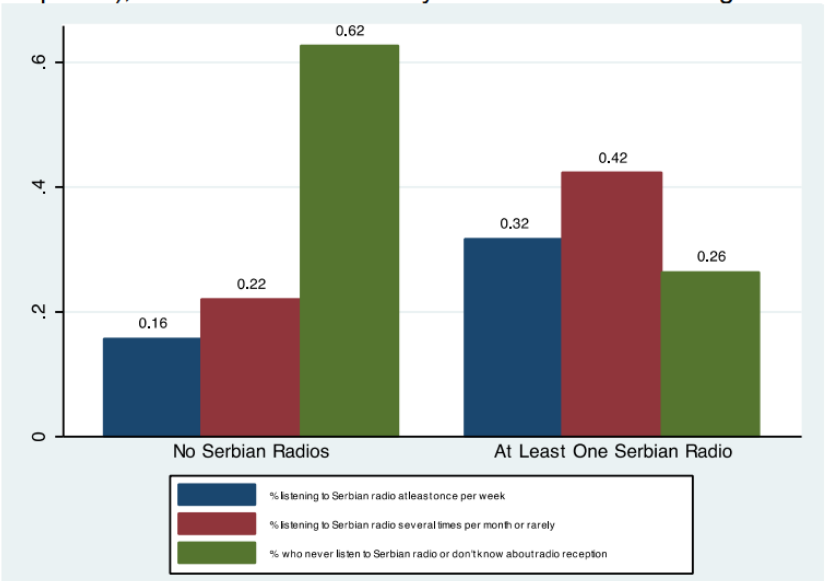


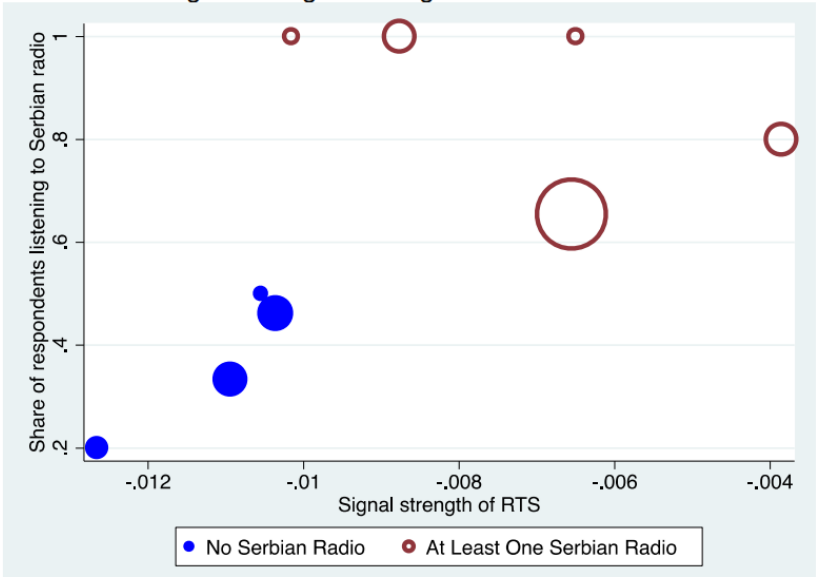
Figure 2b. Reported frequency of listening to Serbian radio (survey response), as function of availability of Serbian radio in village



How often they listen

- Villages with at least one Serbian station: 26 percent never listen, 32 percent listen at least once a week.
- Villages without: 62 percent never listen, 16 percent at least once a week.
- In areas where the signal is available for ordinary radio sets, **78% acknowledge listening occasionally**.
- That sits badly with the extreme view that people only consume media that match their politics.

Figure 2c. Reported incidence of listening to Serbian radio (survey response), as function of signal strength in village.



Why would a Croat tune in?

- Respondents who report listening are also more likely to support concerts by Serbian artists and to know their new songs.
- No such correlation for reported listening to Croatian or Bosnian radio.
- A number of former Socialist Yugoslavian singers, popular among Croats, are banned from Croatia for political reasons.
- “a preference for the entertainment content embedded in the RTS programming may be the main reason for the high exposure to Serbian radio, despite its nationalistic content”.

Second question

Is the content actually inflammatory?

A laboratory experiment in Vukovar

- Croatian students from the region, held at the university of Vukovar.
- Three groups, each hearing a short remix: three news segments separated by two songs.
- The first two news segments are the same for everyone, taken from a Croatian station.
- What differs: the two songs and the third news segment.

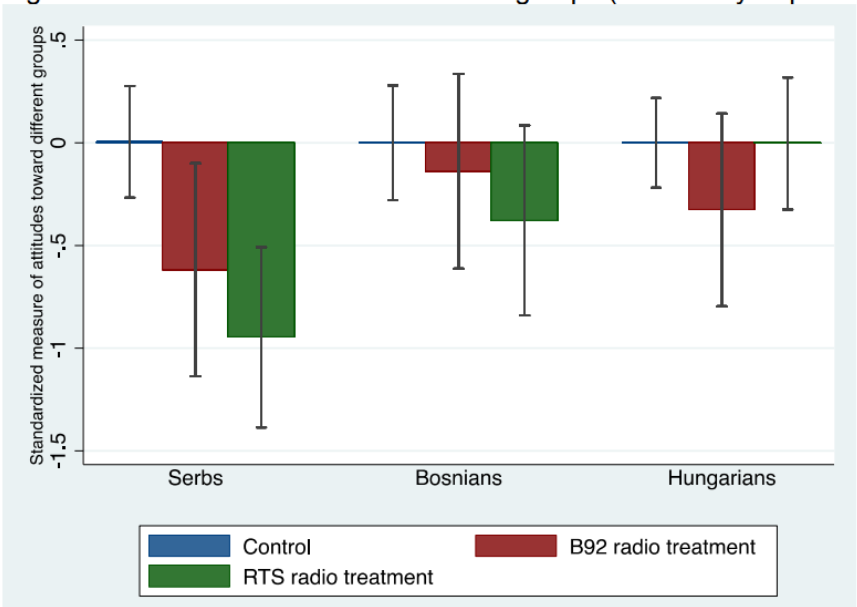
The three arms

- **Control**: Croatian radio, two popular Croatian songs.
- **B92**: the independent Serbian station, two popular Serbian songs.
- **RTS**: the Serbian public station, substantially more nationalistic, another two Serbian songs.
- The Serbian arms carry the same news story, on the Dick Marty report about Kosovo.
- None of the news broadcasts mentions the Serbo-Croatian conflict.

The music, and what is asked afterwards

- Both Serbian arms use modern songs with direct references to Serbian national folklore, easily identifiable by Croats.
- One song in the RTS arm is devoted to a former Serbian paramilitary commander accused of war crimes in Croatia.
- After listening, subjects answer questions about Serbs, Bosnians, Hungarians and Rusini.
- Willingness to work with them, to send children to school with them, to have them as friends, as in-laws, plus a feeling thermometer.

Figure 3. Attitudes toward different ethnic groups (Laboratory Experiment)



Reading the experiment

- Both Serbian treatments push attitudes toward **Serbs** down. The RTS arm pushes them further than B92.
- No movement in attitudes toward Bosnians or Hungarians, as expected.
- B92 is neutral toward Croats; RTS is openly hostile.
- So the channel is not just being reminded of the former war enemy. It is the nationalistic content itself.

Third question

Does it change behavior in the villages?

Is reception as good as random?

- Of 15 non-geographic village characteristics, only education predicts the hand-measured availability.
- Taken together, those characteristics do not predict it. Consistent with reception being idiosyncratic.
- Predicted signal strength is different: it is related to distance to Serbia, population, and the share disabled in the war.
- So for the signal measure, controls and robustness checks carry more weight.

The election result

- Villages with at least one RTS station available: **2.6 percentage points** more vote for the extreme nationalist parties.
- Against a baseline vote share of 7 percentage points in the region.
- So a substantial part of the ultra-nationalist vote in the border region is explained by Serbian radio reception.
- The predicted-signal measure gives the same answer, and adding controls barely moves it.

Figure 4a. Vote share for extreme nationalistic parties in villages with, and without, reception of Serbian radio (controls).

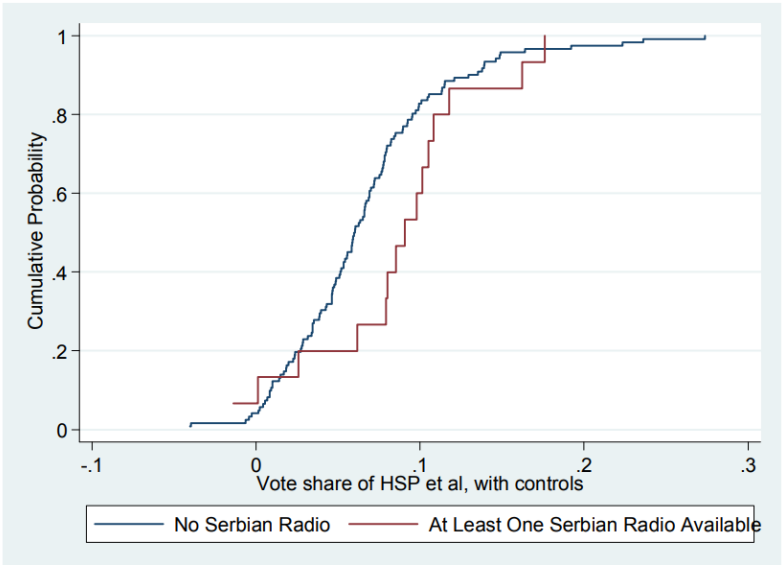


Figure 4b. Map of the vote share for extremely nationalistic parties (with controls) and the measured reception of Serbian radio.

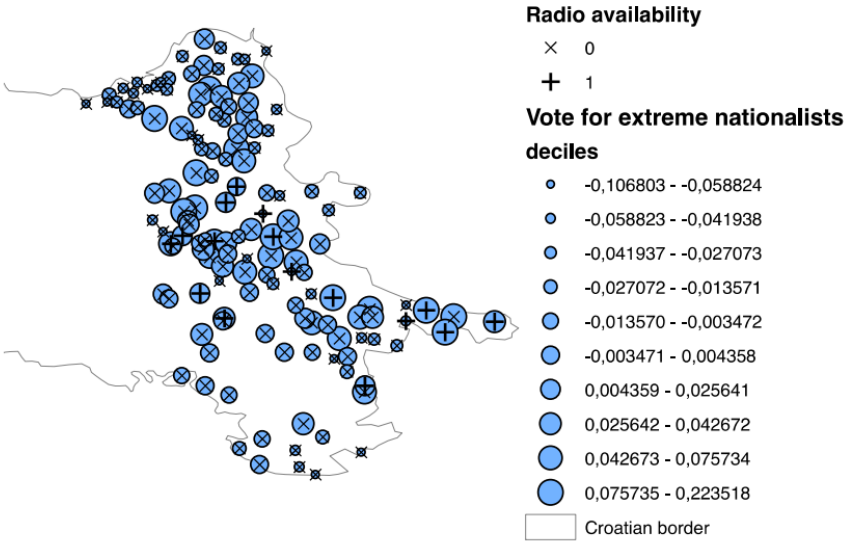
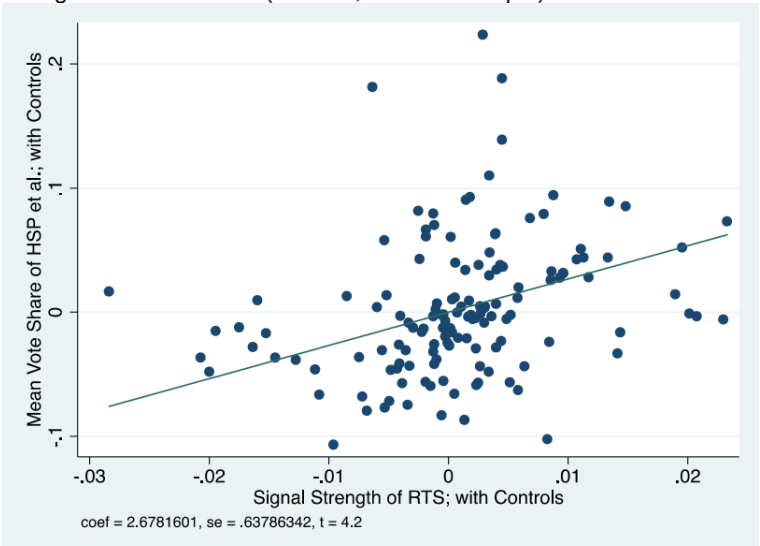


Figure 5. Vote share of extremely nationalistic parties as function of signal strength of Serbian radio (controls, baseline sample).



Notes: The picture shows share of votes for HSP et al. and predicted signal strength of Serbian radio after taking controls into account. Residuals are taken from the specification presented in column (3) of Table 3.

A non-electoral measure: graffiti

- Graffiti coded following the method of the Vukovar NGO Globalpact.
- Examples: “Ubi Srbina” (“Kill a Serb”) and “Srbe na vrbe” (“Hang a Serb on a willow”).
- In villages with Serbian radio, the probability of ethnically offensive graffiti is **35 to 40 percentage points** higher.
- That is roughly double the rate in villages without it.

A radio that should do nothing

- Hungarian radio also crosses into the region, and very few Croats speak Hungarian.
- It has no effect on nationalist voting, and controlling for it leaves the Serbian effect intact.
- Controlling for the signal of the two major Croatian networks changes nothing either.
- So the result is not about villages that simply happen to get good radio reception in general.

What the paper documents

- A decade after the shooting stopped, Serbian public radio still crossed the border and was heard by Croats.
- Entertainment pulled them in; political content came with it.
- Where it was received: more ultra-nationalist voting, more offensive graffiti, a more polarized electorate.
- "nation-building efforts implicit in the nationalistic content of the group-specific media (in our case, the Serbian radio) can have important negative spillovers on the persistence of peace."

Radio and the rise of the Nazis

Adena, Enikolopov, Petrova,
Santarosa and Zhuravskaya (2015)

The question

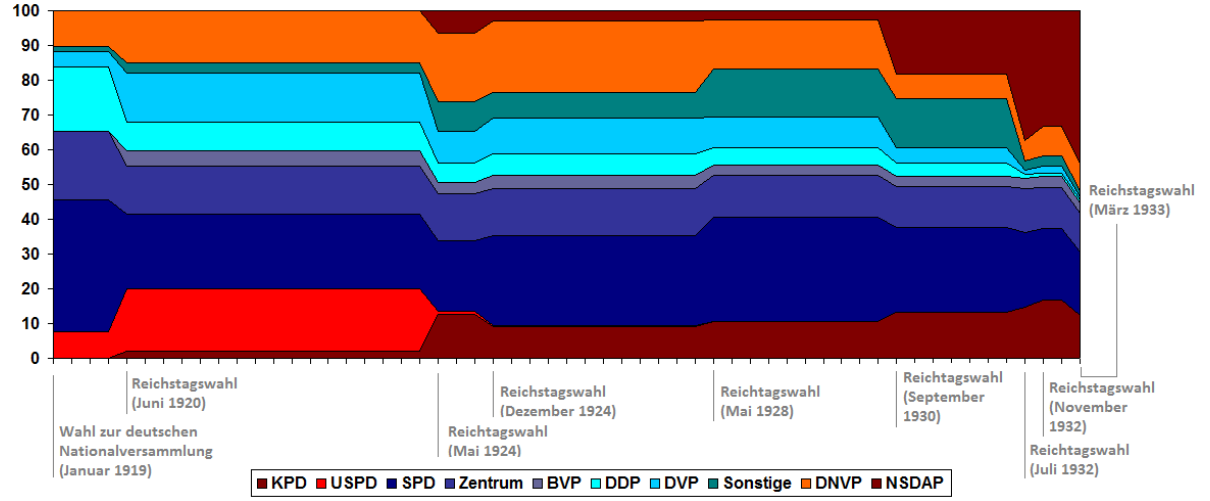
- How do the media affect public support for democratic institutions in a **fragile democracy**?
- What role do they play once the dictatorship is in place?
- The rise of the Third Reich was the most prominent collapse of a democracy without a military coup.
- Same team of authors, overlapping with the Croatia paper, same signal-propagation toolkit.

Goebbels believed it worked

- At the Berlin radio exhibition, August 1933: “It would not have been possible for us to take power or to use it in the ways we have without the radio.”
- In his diary, on the weeks before the last competitive election: “Now it will be easy to carry on the fight, for we can call on all the resources of the State. Radio and press are at our disposal. We shall stage a masterpiece of propaganda”.

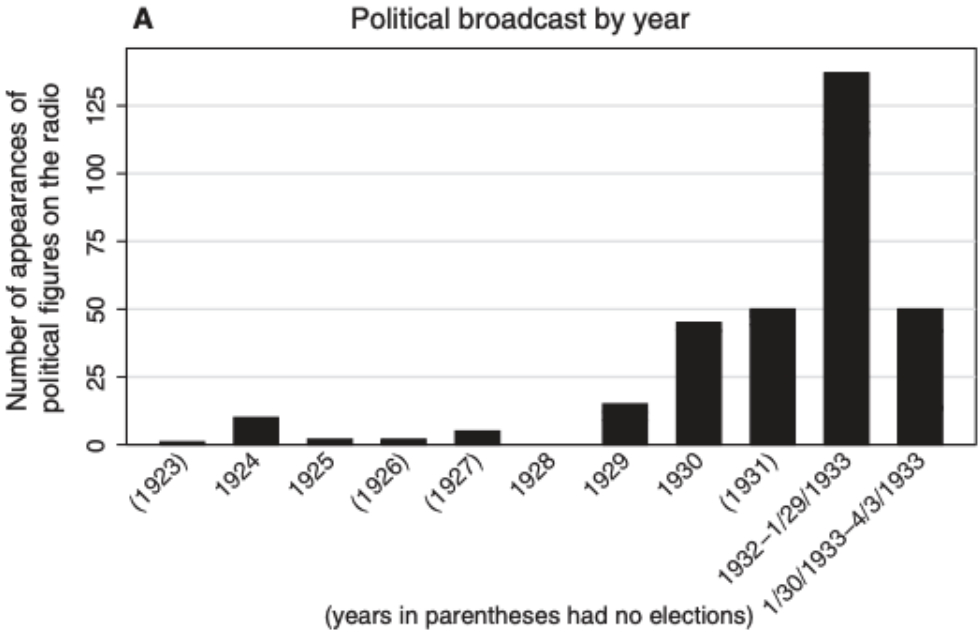
Weimar politics

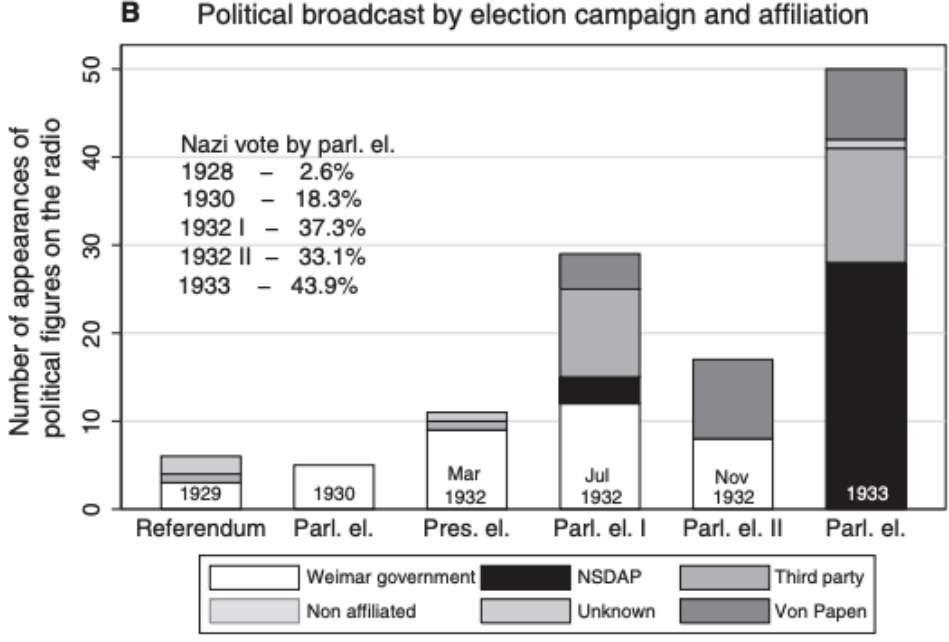
- Parliamentary democracy from 1919, governed by a coalition of centrist parties led by the SPD.
- It survived hyperinflation and held a stable majority until 1930.
- The Depression, the recall of US short-term loans and the reparations burden broke it.
- NSDAP vote: 2.6% in 1928, 18.3% in 1930, 37.3% in July 1932, 33.1% in November 1932.
- Hitler appointed chancellor on January 30, 1933; the NSDAP then took 43.9% in March.


100
90
80
70
60
50
40
30
20
10
0
Wahl zur deutschen Nationalversammlung (Januar 1919)
Reichstagswahl (Juni 1920)
Reichstagswahl (Mai 1924)
Reichstagswahl (Dezember 1924)
Reichtagswahl (Mai 1928)
Reichtagswahl (September 1930)
Reichstagswahl (Juli 1932)
Reichstagswahl (November 1932)
Reichstagswahl (März 1933)
KPD
USPD
SPD
Zentrum
BVP
DDP
DVP
Sonstige
DNVP
NSDAP

Radio content changed twice

- Until 1928: deliberately apolitical. A 1926 regulation forbade partisan content; no campaign content aired in 1928.
- **1929**: right-wing parties organize a referendum to repudiate the reparations of the Treaty of Versailles.
- The government answers with an intensive campaign against it. Radio turns pro-government and pro-democratic.
- The Nazis and the communists are denied airtime, unlike the other parties.
- **January 30, 1933**: Hitler takes the executive powers, radio among them. Content turns pro-Nazi.


A
Political broadcast by year
Number of appearances of political figures on the radio
0
25
50
75
100
125
(1923)
1924
1925
(1926)
(1927)
1928
1929
1930
(1931)
1932–1/29/1933
1/30/1933–4/3/1933
(years in parentheses had no elections)


B
Political broadcast by election campaign and affiliation
Number of appearances of political figures on the radio
0
10
20
30
40
50
Nazi vote by parl. el.
1928 – 2.6%
1930 – 18.3%
1932 I – 37.3%
1932 II – 33.1%
1933 – 43.9%
1929
1930
Mar 1932
Jul 1932
Nov 1932
1933
Referendum
Parl. el.
Pres. el.
Parl. el. I
Parl. el. II
Parl. el.
Weimar government
NSDAP
Third party
Non affiliated
Unknown
Von Papen

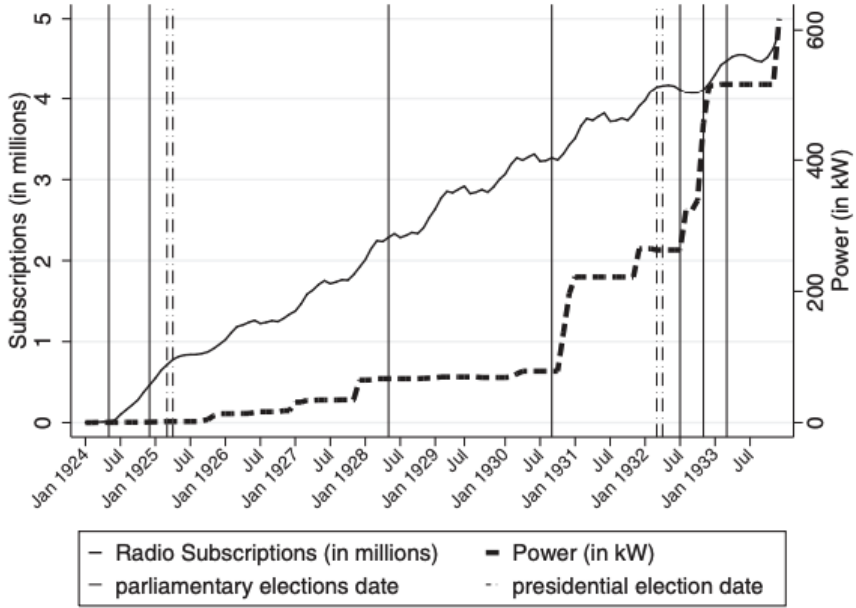
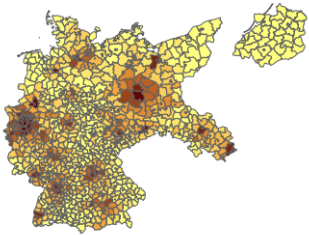
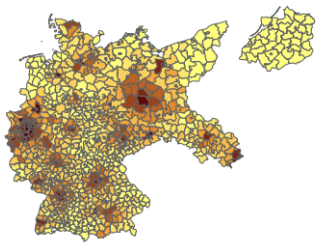

Subscriptions (in millions)
0
1
2
3
4
5
Power (in kW)
0
200
400
600
Jan 1924
Jul
Jan 1925
Jul
Jan 1926
Jul
Jan 1927
Jul
Jan 1928
Jul
Jan 1929
Jul
Jan 1930
Jul
Jan 1931
Jul
Jan 1932
Jul
Jan 1933
Jul
Radio Subscriptions (in millions)
Power (in kW)
parliamentary elections date
presidential election date

Figure A1. Radio Signal Strength in Germany. Raw signal strength measure.

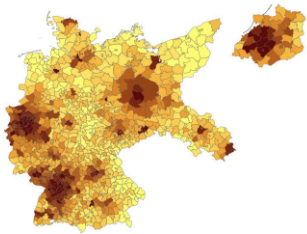
A. May 1928



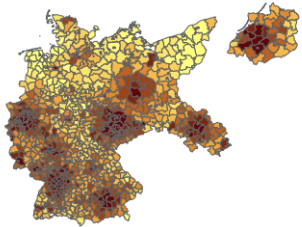
B. September 1930



B. July 1932



D. March 1933



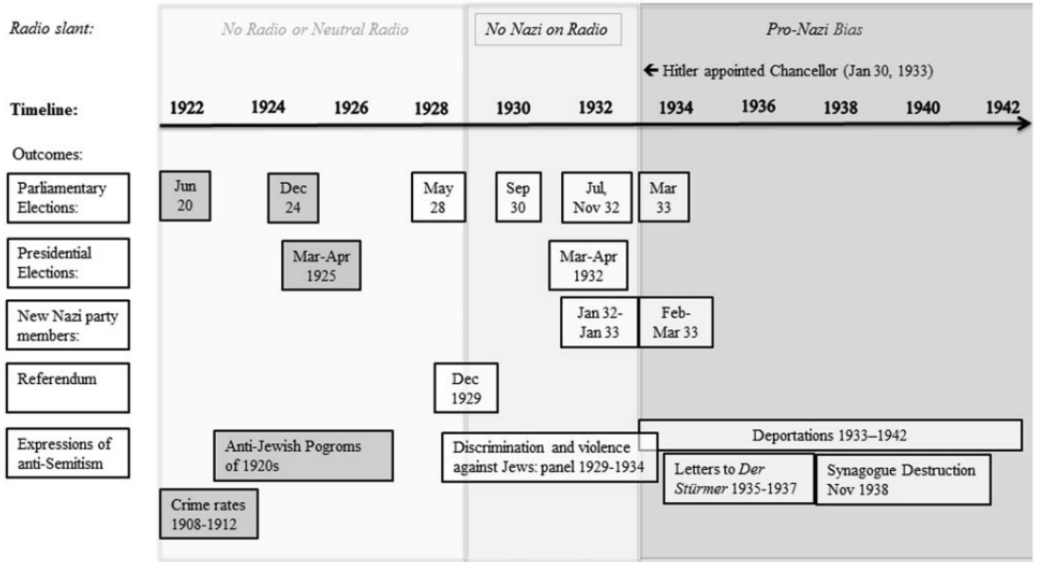


FIGURE III

The Timing of the Measurement of Outcome Variables and of the Change in Radio Content

What is measured, and when

- Top band: the slant of the radio. No radio or neutral, then no Nazi on radio, then pro-Nazi from 1933.
- Outcomes below: parliamentary and presidential elections, new Nazi party members, the 1929 referendum.
- Expressions of antisemitism: discrimination and violence 1929 to 1934, deportations 1933 to 1942, letters to a Nazi newspaper, synagogue destruction in November 1938.
- Crime rates from 1908 to 1912 and the pogroms of the 1920s sit before the radio: they are the checks that should find nothing.

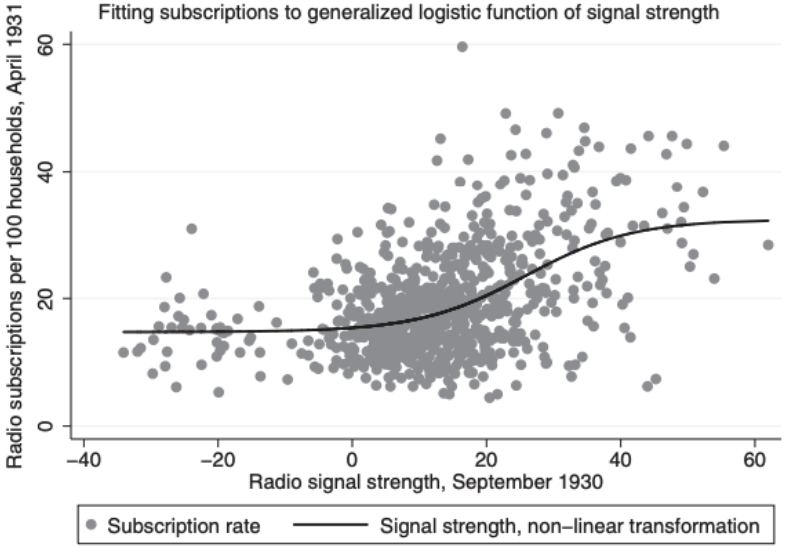


FIGURE IV

Radio Subscriptions and Signal Strength, 1931 (No Controls)

TABLE I

RADIO SUBSCRIPTIONS AND SIGNAL STRENGTH

	(1)	(2)	(3)	(4)
	Share of households with radio subscription at a given date			
Date for the subscription rate	Apr 1931	Apr 1932	Apr 1932	Apr 1933
Date for the signal strength	Sep 1930	Jul 1932	Nov 1932	Mar 1933
Panel A				
Radio signal strength	0.218***	0.175***	0.161***	0.163***
	[0.030]	[0.028]	[0.031]	[0.036]
Distance to the nearest big city, city dummy, altitude	Yes	Yes	Yes	Yes
Socioeconomic controls	Yes	Yes	Yes	Yes
Population, fifth-order polynomial	Yes	Yes	Yes	Yes
Voting controls, 1924	Yes	Yes	Yes	Yes
Region fixed effects	Yes	Yes	Yes	Yes
Observations	858	883	883	885
R^2	0.576	0.531	0.526	0.514
F-stat for the signal strength variable	50.19	36.13	24.53	20.30
Panel B				
Radio signal strength, nonlinear transformation	0.671***	0.683***	0.656***	0.602***
	[0.096]	[0.100]	[0.120]	[0.123]
Distance to the nearest big city, city dummy, altitude	Yes	Yes	Yes	Yes
Socioeconomic controls	Yes	Yes	Yes	Yes
Population, fifth-order polynomial	Yes	Yes	Yes	Yes
Voting controls, 1924	Yes	Yes	Yes	Yes
Region fixed effects	Yes	Yes	Yes	Yes
Observations	858	883	883	885
R^2	0.579	0.533	0.525	0.513
F-stat for the signal strength variable	46.62	42.41	26.38	23.05

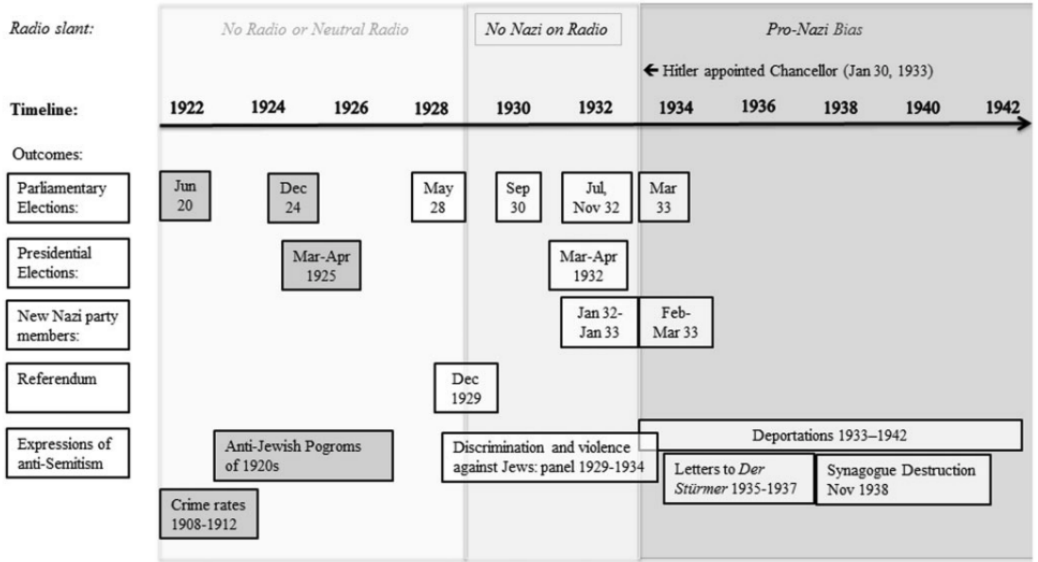


FIGURE III

The Timing of the Measurement of Outcome Variables and of the Change in Radio Content

TABLE III

Radio Availability and Voting for the Nazis: District Fixed Effects

	(1)	(2)	(3)	(4)	(5)	(6)
	Nazi vote share				Change in the Nazi vote share	
Time span of the sample	September 1930, July 1932, and November 1932		All parliamentary elections 1928–1933, combined		Between September 1930 and November 1932	
Specification	Panel: OLS	Panel: OLS	Panel: OLS	Panel: OLS	First differences: OLS	First differences: IV
Radio signal strength	−0.079**					
	[0.034]					
Radio signal strength, nonlinear transformation		−0.227**				
		[0.088]				
Radio signal strength × pro-Nazi slant			0.028***			
			[0.007]			
Nonlinear transformation of signal strength × pro-Nazi slant				0.123***		
				[0.027]		
Change in subscription rate between April 1931 and April 1932					−0.104	−2.127**
					[0.098]	[0.873]
Baseline controls, interacted with time fixed effects	Yes	Yes	Yes	Yes	No	No
Baseline controls	No	No	No	No	Yes	Yes
District fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	No	No
Observations	2,836	2,836	4,713	4,713	827	827
Number of districts	959	959	959	959	827	827
R^2	0.939	0.939	0.972	0.972	0.658	
F-statistics for instrumental variable						26.44

Notes. Standard errors clustered at the region level in parentheses. * $p < .1$, ** $p < .05$, *** $p < .01$. Pro-Nazi slant equals 0 in 1928, -1 between 1930 and 1932, +1 in 1933. Controls include fifth-order polynomial of population, share of Jewish and Catholic population, shares of blue- and white-collar workers in 1925, city dummy, number of war participants per 1,000, number of welfare recipients per 1,000, number of renters of social housing per 1,000, log of average property tax, altitude, share of unemployed and partially employed, distance to the nearest city with population over 50k, turnout and vote shares of DNVP, NSFP, SPD, and Zentrum in December 1924. Number of observations changes between elections because of redistricting. In column (8) change in radio subscription rate is instrumented using change in nonlinear transformation of the signal strength.

Pro-Weimar radio slowed the Nazis

- 8.2 percentage points more predicted radio subscription in 1930 (one standard deviation of the actual rate) $\rightarrow$ a **1.8 pp smaller** rise in the Nazi vote between 1928 and 1930.
- Using actual subscriptions rather than predicted ones, the same exercise gives 2.9 pp.
- Tracking the change in subscriptions district by district between September 1930 and November 1932: a fall of 4.9 pp.
- Counterfactual with no radio during the September 1930 campaign: the Nazis get 22.3% instead of 18.2%.
- The SPD got 24.5%. That gap mattered for bargaining over the chancellorship already in 1930.

TABLE IV

RADIO AND AN INCREASE IN NAZI VOTE SHARE

	(1)	(2)	(3)	(4)
	Change in vote share of the Nazi party since previous elections			
Election dates	Sep 1930 (change from May 1928)		Mar 1933 (change from Nov 1932)	
Panel A: reduced-form estimation				
Radio signal strength	−0.061***		0.045**	
	[0.022]		[0.020]	
Radio signal strength, nonlinear transformation		−0.217***		0.128*
		[0.071]		[0.071]
Region fixed effects	Yes	Yes	Yes	Yes
Baseline controls	Yes	Yes	Yes	Yes
Observations	958	958	918	918
Panel B: OLS and IV results				
Date for the subscription rate variable	Apr 1931		Apr 1933	
Specification	OLS	IV	OLS	IV
Radio subscription rate, %	−0.086*	−0.347***	0.032*	0.222*
	[0.045]	[0.095]	[0.017]	[0.117]
Region fixed effects	Yes	Yes	Yes	Yes
Baseline controls	Yes	Yes	Yes	Yes
Observations	857	855	853	851
F-statistic for the exclusion of the instrument		50.19		20.30
Panel C: Altonji-Elder-Taber tests				
Index of observables (predicted signal strength)	−0.037		−0.047	
	[0.070]		[0.049]	
Index of observables (predicted nonlinear signal strength)		0.014		−0.348*
		[0.210]		[0.200]
Population, fifth-order polynomial	Yes	Yes	Yes	Yes
Observations	958	958	918	918
R^2	0.432	0.432	0.573	0.576

The sign flips in 1933

- Left columns: change in the Nazi vote from May 1928 to September 1930. Negative in signal strength.
- Right columns: change from November 1932 to March 1933. Positive.
- Same districts, same measure of exposure, opposite sign. What changed is who controlled the content.
- Each district is compared with itself over time, so fixed district traits cannot drive it.
- Fixing the signal map at earlier years and re-running gives the same flip, so it is not the spread of radio.

Nazi radio, five weeks of it

- The same 8.2 pp increase in predicted subscription in 1933 $\rightarrow$ a **1 pp larger** rise in the Nazi vote between November 1932 and March 1933.
- With actual subscriptions rather than predicted ones: 2.0 pp.
- Counterfactual with the radio switched off in January 1933: 41.0% instead of 43.9%.
- With their coalition partner's 8%, that would not have been a majority of seats.
- About a quarter of what the NSDAP gained in those four months, from five weeks of broadcasting.

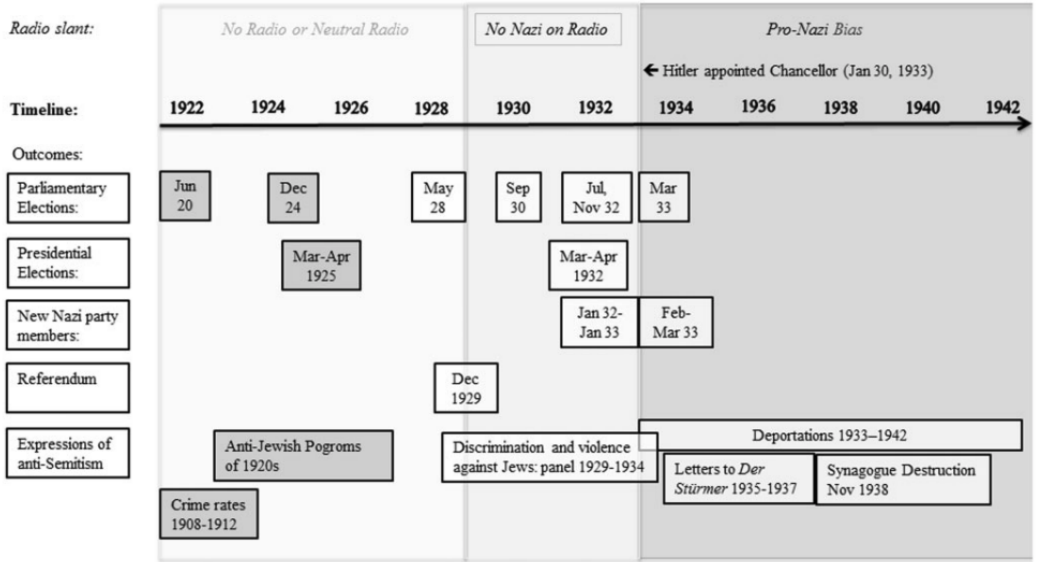


FIGURE III

The Timing of the Measurement of Outcome Variables and of the Change in Radio Content

TABLE V

RADIO AVAILABILITY AND VOTING IN ANTI–VERSAILLES TREATY REFERENDUM AND APRIL 1932 PRESIDENTIAL ELECTIONS

	(1)	(2)	(5)	(6)	(7)	(8)
	Referendum on the Law against the Enslavement of the German People		1932 presidential elections, 1st round			
	Voted "yes" in the referendum (share of eligible voters)		Von Hindenburg vote share		Hitler vote share	
Panel A: radio availability and other voting outcomes.						
Radio signal strength	−0.065**		0.054**		−0.049	
	[0.031]		[0.026]		[0.036]	
Radio signal strength, non-linear transformation		−0.244**		0.194		−0.240
		[0.095]		[0.118]		[0.149]
Nazi party vote share in 1928	0.647***	0.639***	−0.457***	−0.456***	0.584***	0.584***
	[0.090]	[0.090]	[0.070]	[0.071]	[0.064]	[0.064]
Region fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Baseline controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	949	949	952	952	952	952
R^2	0.845	0.846	0.881	0.881	0.876	0.876
Panel B: Altonji-Elder-Taber tests						
Index of observables (predicted signal strength)	−0.138		−0.101		−0.161	
	[0.133]		[0.174]		[0.157]	
Index of observables (predicted nonlinear transformation of signal strength)		−0.525		−1.291*		−0.113
		[0.393]		[0.694]		[0.634]
Nazi party vote share in 1928	1.276***	1.283***	−1.490***	−1.509***	1.490***	1.526***
	[0.091]	[0.090]	[0.129]	[0.119]	[0.128]	[0.123]
Population, fifth-order polynomial	Yes	Yes	Yes	Yes	Yes	Yes
Observations	949	949	952	952	952	952
R^2	0.229	0.231	0.202	0.213	0.227	0.225

TABLE VII

Radio Availability and Nazi Party Membership

Time period	(1)	(2)	(3)	(4)	(5)	(6)
	Log of the number of new party members of NSDAP					
	Jan 1932–Jan 1933		Feb–May 1933			
Panel A: cross-sectional estimates						
Radio signal strength	0.0032		0.0053**		0.0052**	
	[0.0020]		[0.0020]		[0.0020]	
Radio signal strength, nonlinear transformation		0.0006		0.0201**		0.0203**
		[0.0099]		[0.0077]		[0.0079]
Log of new party membership in 01/1932–01/1933					0.0544*	0.0579*
					[0.0292]	[0.0289]
Region fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Baseline controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	613	613	613	613	613	613
R^2	0.677	0.676	0.344	0.345	0.348	0.349
Panel B: Altonji-Elder-Taber tests						
Index of observables (predicted signal strength)	0.0026		0.0013		0.0016	
	[0.006]		[0.004]		[0.004]	
Index of observables (predicted nonlinear signal strength)		0.0221		0.0077		0.0080
		[0.029]		[0.017]		[0.017]
Log of new party membership in 01/1932–01/1933					0.0484*	0.0480*
					[0.027]	[0.027]
Population, fifth-order polynomial	Yes	Yes	Yes	Yes	Yes	Yes
Observations	613	613	613	613	613	613
R^2	0.587	0.587	0.288	0.288	0.292	0.292

Recruiting party members

- New NSDAP members by city, from the party membership cards.
- In January 1932 to January 1933, with pro-Weimar radio: nothing. Those who joined then were core supporters.
- In February to May 1933, after the takeover of the radio: a clear positive relationship with signal strength.
- The implied magnitude: **58% of new members** who joined in the first two months of Nazi control of the broadcast were persuaded by radio propaganda.
- The party stopped accepting members in May 1933 because of the flood of applicants.

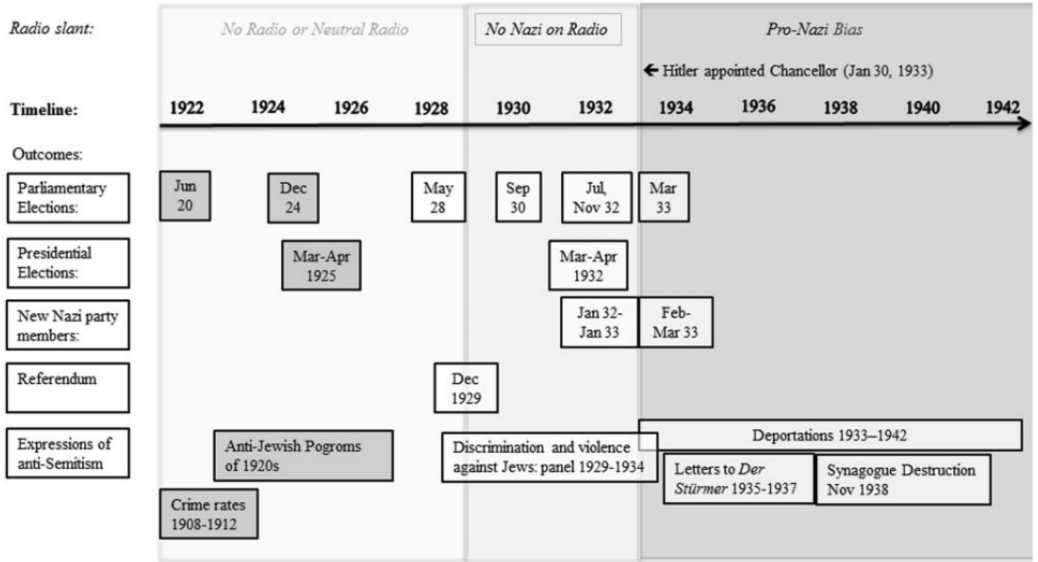


FIGURE III

The Timing of the Measurement of Outcome Variables and of the Change in Radio Content

TABLE VIII

RADIO AVAILABILITY AND ANTI-SEMITISM

Model	(1) Log(deportations before 1942) OLS	(2) Letters to *Der Stürmer* ML	(3) Attacks on synagogues OLS
Sample	All cities	All cities	All cities with synagogues in 1933
Panel A: Baseline results.			
Radio signal strength in 1937	0.0187**	0.0194***	0.0001
	[0.0069]	[0.0068]	[0.0014]
Log (population)	0.3447***	0.3900***	−0.0097
	[0.0803]	[0.0634]	[0.0141]
Altitude	−0.0009*	−0.0008	−0.0001
	[0.0005]	[0.0005]	[0.0002]
City located at navigable river	0.3693***	0.7504***	0.0467**
	[0.0683]	[0.1191]	[0.0189]
Socioeconomic controls	Yes	Yes	Yes
Voting controls, 1924	Yes	Yes	Yes
Region fixed effects	Yes	Yes	Yes
Observations	1,325	1,391	1,134
R^2	0.385		0.103
Panel B: Altonji-Elder-Taber style test			
Index of observables (Prediction of signal strength)	−0.0084	0.0245	0.0103**
	[0.0263]	[0.0221]	[0.0045]
Population	Yes	Yes	Yes
Geographic variables	Yes	Yes	Yes
Region fixed effects	Yes	Yes	Yes
Observations	1,325	1,391	1,133
R^2	0.197		0.064

Notes. Results of OLS estimation in columns (1) and (3); results of negative binomial maximum likelihood estimation in column (2). Standard errors are clustered by region (Wahlkreis). * $p < .1$, ** $p < .05$, *** $p < .01$. Unit of observation is city in Voigtländer and Voth (2012) sample. Socioeconomic controls include share of Jewish and Catholic population, shares of blue- and white-collar workers in 1925, number of war participants per 1,000, number of welfare recipients per 1,000, number of renters of social housing per 1,000, log of average property. Voting controls include voter turnout and vote shares of DNVP, NSFP, SPD, and Zentrum in December 1924. Geographic controls include altitude and dummy for being located on a navigable river.

After power was fully consolidated

- Elections stop being informative in a dictatorship, so the outcomes change.
- Jews deported to concentration camps before 1942, mostly following denunciations by neighbors and local officials.
- Open antisemitic letters to the Nazi newspaper *Der Stürmer*, 1935 to 1938. Attacks on synagogues in November 1938.
- Stronger signal in 1937 (one standard deviation) $\rightarrow$ **21% more deportations** (mean 11.7 per city) and **22% more letters** (mean 1.76, so 0.38 more).
- No effect on synagogue attacks.

TABLE IX

Radio Availability and Anti-Semitism, the Role of Historic Predispositions

Model	(1) Log(deportations before 1942) OLS	(2) Letters to *Der Stürmer* ML	(3) Attacks on synagogues OLS
Panel A: interactions with pogroms			
Sample	Subsample of cities with historical Jewish settlement		
Pogroms in 1349 * Radio signal strength, demeaned	0.048***	0.073***	0.003
	[0.016]	[0.016]	[0.007]
Radio signal strength, 1937	−0.032*	−0.058***	−0.004
	[0.017]	[0.016]	[0.007]
Pogroms in 1349	0.701***	0.601***	0.160**
	[0.171]	[0.227]	[0.065]
All baseline controls	Yes	Yes	Yes
Region fixed effects	Yes	Yes	Yes
Observations	296	319	274
R^2	0.658		0.253

Propaganda meets prior beliefs

- Cities are split by historical antisemitism: whether they had anti-Jewish pogroms during the Black Death of 1349.
- Where the pogroms happened, the effect of radio is 2.6 to 3.8 times larger than in the average city.
- Same pattern using the 1924 vote for the extreme nationalist NSFP, and using land inequality in 1895.
- **Where they did not happen, the sign flips**: 28% fewer deportations and 45% fewer letters.

What the paper documents

- Mass media were a safeguard against the fall of an unconsolidated democracy, and then a tool in its collapse.
- What flipped the sign was control over content, plus whether extremists were barred from the air.
- Under the dictatorship, radio raised popular participation in the regime's worst policies.
- But only where the audience did not disagree a priori. Elsewhere it produced resistance.

Historical Antisemitism, Ethnic Specialization, and Financial Development

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Historically, European Jews have specialized in financial services while being the victims of antisemitism. We find that the present-day demand for finance is lower in German counties where historical antisemitism was higher, compared to otherwise similar counties. Households in counties with high historical antisemitism have similar saving rates but invest less in stocks, hold lower saving deposits, and are less likely to get a mortgage to finance homeownership after controlling for wealth and a rich set of current and historical covariates. Present-day antisemitism and supply-side forces do not fully explain the results. Households in counties where historical antisemitism was higher distrust the financial sector more—a potential cultural externality of historical antisemitism that reduces wealth accumulation in the long run.

Key words: Cultural economics, Cultural finance, Intergenerational transmission of norms, Stereotypes, Household finance, History & finance.

JEL Codes: D91, G11, J15, N90, Z10, Z12

Taking stock

Two settings, one source of variation

- Both read the same physics: transmitter power, distance, terrain, then the Irregular Terrain Model.
- Croatia: reception varies village by village, in a jagged way, near the Serbian border.
- Germany in the 1930s: reception varies across districts and over time as transmitters open.
- In each case the listener did not choose the signal, and often was not the intended audience.

What determines whether media move people

- **Content**: the same German radio, in the same districts, pushed against the Nazis and then for them.
- **Control**: who holds the transmitters, and whether extremists are allowed on the air.
- **Predisposition**: propaganda worked where the audience half believed it and backfired where it did not.
- Croatia adds a fourth: the audience does not have to be the target, and entertainment can carry the politics.

Where else this shows up

- Yanagizawa-Drott: hate speech on the radio and casualties in the Rwandan genocide of 1994.
- Paluck and Green: an appeasing educational broadcast in post-conflict Rwanda, run as a field experiment.
- Gentzkow and Shapiro: media in Arabic countries and anti-American sentiment.
- La Ferrara, Chong and Duryea: exposure to television soap operas and fertility in Brazil.
- For discussion: in Brazil, what would play the role of the terrain here?